The effect of policy changes and economic market shocks on abortion rates in Hungary from 2001 to 2024

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1. Research Problem

The discussion of abortion is a highly personal and sensitive topic, which has gained wide political attention in recent years not only in the US (Paula M. et al., 2023) but in Central and Eastern Europe, like Poland (A. Kulczycki, 2023) and Hungary (I. Szalma et al., 2024) as well. While the certain governments tend to make a purely political and moral question out of this topic, scientists agree that restricting the rights of women for abortion does not necessarily decrease the number of abortions but significantly increases the health risk of the women and results in higher maternal morbidity rates, and results in an especially large increase in the case of worse off women from minorities (Silpa et al., 2025, A. Gemill et al., 2024). Instead of legally banning abortion, better understating what determinants lead to an increased probability of abortion (J Čihák et al., 2025, Szabó-Morvai et al., 2023) and directly addressing them, would be more beneficial for both policy makers and women.

However, before examining the issue from a policy perspective, it is worth considering why and in what direction abortion rates can change. Several studies show that critical life stages and events (e.g. economic uncertainty) can be linked to pregnancy termination. (Minkus et al. 2021). For this reason, it is worth examining the effects of global events, such as the 2008-2009 Global Financial Crisis, which can trigger these changes (Gonzalez, F., & Quast, T. 2022). In addition, this figure may also depend on the availability of safe abortion services. For example, during the COVID-19 pandemic, many people were unable to access adequate healthcare due to the strain on the healthcare system and uncertainty, and several governments also tightened access to abortion, which may be reflected in the figures for this period (Moreau C. et al. 2021, Caruana-Finkel L., 2020).

In case of the Hungarian policy environment, the topic of abortion has gained high political attention from the start of the second Orbán government, which started in 2010. There was a strong shift in the orientation of Hungarian politics towards the aim to promote child-bearing, since Hungary experienced a drastic decrease in the willingness to have children in recent the previous decades (L. Csaba et al., 2024). Therefore the government introduced several policies with the hope to reach the replacement-level fertility (I. Szalma et al., 2024). On one hand the government offered a wide range of family support elements, like tax benefits, favorable financial schemes, loans and loan waivers for families with only children or in the case of having children, subsidies to support the growth of wealth, labor market benefits, and child welfare measures (L. Csaba et al., 2024). On the other hand the government also introduced

stricter abortion policies in Hungary, while still keeping women's access to abortion as set out it in the Act LXXIX of 1992 on the Protection of Fetal Life. In 2011 the following sentence was added to the Fundamental Law of Hungary: "From the moment of conception, the fetus is entitled to protection, respect and support as provided by law" (Act CCXI of 2011). Another crucial event took place in 2022, when the "fetal heartbeat" amendment was introduced, which requires women to listen to the heartbeat of the fetus before the abortion procedure (I. Szalma et al., 2024).

Previous studies have also elaborated on how the state of the job market effects the number of abortions, however there is no general consensus in their results. Blank et al., 1996 found a strong, positive correlation between unemployment and abortion, in contrast F. Gonzalez et al., 2022 showed procyclical association (increase in unemployment is associated with a decrease in abortion rate), while several other studies failed to find a relationship between fertility and unemployment rate (Dehejia et al., 2004). Szabó-Morvai et al., 2023 were the first to show that different types of employments shocks - like firm closures and mass layoffs - play a role in the number of induced abortions and find that in case there are dismissal protection for pregnant women - which means that pregnant women cannot fired – and high maternity benefits, women tend to keep their pregnancies, while if they are unavailable the probability of abortions increase.

2. Research Question

Focusing on Hungary, our research wants to examine in great detail the various regulatory changes that impacted number of safe and legal abortions. In addition we want to analyze what factors and determinants drive abortion numbers. We want to focus on linking abortion rates to labor markets and analyze whether specific labor market shocks, related to deployment, wages and career have a significant effect on the abortion rates. We would also like to investigate whether various major global economic and other significant events (such as the Global Financial Crisis of 2008-2009, COVID-19, etc.) had an impact on abortion rates.

We aim to examine the following questions:

- Did the rate of abortions change significantly and permanently around 2011–2012, coinciding with the public debate and legislative changes concerning the Fundamental Law?
- What was the immediate and long-term effect of the introduction of the "fetal heartbeat regulation" in 2022 on the monthly number of registered terminations of pregnancy in Hungary?
- How does an increase in the county-level unemployment rate influence the abortion rates across different demographic groups (e.g., age, educational attainment)?
- What is the effect of county-level changes in real average wages on the number of abortions in Hungary between 2001 and 2024?
- Among the various labor market shocks (unemployment, wages, firm closure, job transfer), which one possesses the most significant predictive power regarding changes in the abortion rate?
- Did the onset of the Global Financial Crisis 2008-2009 (GFC) cause an immediate and significant increase in the monthly abortion rate, reflecting acute economic anxiety?
- How did the lockdowns and economic uncertainty resulting from the COVID-19 pandemic (separate from the labor market indicators) influence the number of terminations during the 2020–2021 period?

3. Description of Data Collection

We use monthly data on demographics (age, county of residence), job shock indicators (unemployment, job transfer, firm closure, wage) and the number of pregnancy terminations (abortion rates) across Hungary from 2001 to 2024. We also take into account the event of abortion regulation changes (Fundamental Law, fetal heartbeat regulation) and global shocks, like the GFC and COVID-19. We collect data from various sources; Databank of the Centre for Economic and Regional Studies (ELTE KRTK), Eurostat, and Hungarian Central Statistical Office (KSH). We take into account and control for the seasonality of the monthly abortion data (Wesselink et al., 2022).

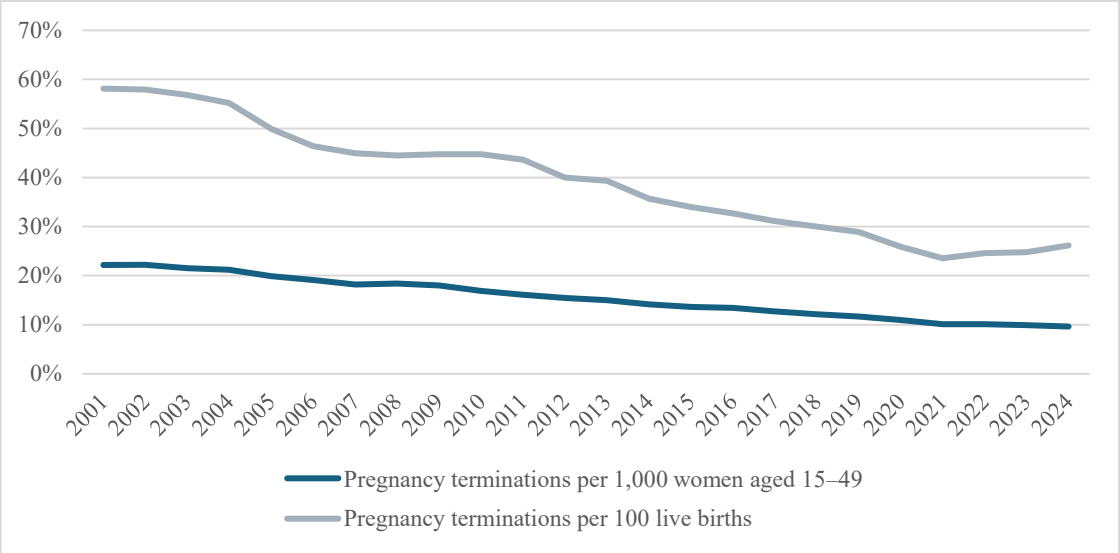


Figure 1: Pregnancy terminations from 2001 to 2024

4. Methodology

In order to study the effects of job market, policy changes and economic market shocks we employ Ordinary Least Squares (OLS) regression in line with previous studies (Čihák et al., 2025, Ahmed and Ray, 2014, Blank et al., 1996). The choice between the fixed effect and the random effects OLS model will be decided by the *p* value determined by the Hausman test, however the use of the fixed effect model is expected. The linear regression equation is the following: the dependent variable represents the abortion rate at a given year *t* in Hungary, while the explanatory variables control for the abortion policy changes, labor market and economic market shocks, where we interact unemployment rate with the fixed year effect (F. Gonzalez et al., 2022). In our model we control for autocorrelation and heteroskedasticity. In order to confirm our results, we also employ event study estimates, to check the robustness of our findings.

5. Expected Outcomes

Our analysis aims to quantify the distinct impact of the strengthening of the abortion policy, job market related shocks, and the major economic crises on Hungary's abortion rates. We expect our models to generate the following key results.

We predict that the 2022 Fetal Heartbeat Regulation will manifest as a clear immediate drop in the abortion rate, captured by a significantly negative level shift in our time-series models. However, we anticipate this reduction will be short-lived, suggesting the policy primarily creates a temporary barrier rather than altering the long-term trend. We expect that COVID-19

pandemic causes an immediate decrease in the number of abortions, reflecting on the overburdened healthcare system and the restrictive measures imposed due to the emergency situation (Moreau C. et al., 2021).

We expect strong confirmation that economic distress drives abortion rates. We anticipate that rising county-level unemployment and, even more critically, declines in real wages will correlate significantly and positively with the number of abortions. In addition, we also expect that the financial uncertainty following the GFC will also lead to an increase in the number of abortions. This outcome would highlight that threats to household financial stability - particularly those related to income - are the most powerful and reliable predictors of abortion rate fluctuations across Hungarian counties.

6. Impact and Further Research

This study is relevant for the government and policy makers. Unraveling the determinants of abortion and therefore gaining a deeper understanding in such action could lead to developing more effective policies that aim to maximize the number of births, while minimizing the number of abortions than restricting the rights of pregnancy terminations. Making it more difficult to get the necessary medical treatment or stigmatizing abortion will no longer be the only tool in their basket. The research aims to show how the number of abortions depends on many factors and how various political efforts influence or fail to influence this. The political effort to reduce the number of abortions to the lowest possible level is understandable, but it is important to see the underlying cause and to achieve this not through restrictions or strong pressure, but through appropriate education and genuine help. Research can be conducted to uncover the real reasons behind unwanted pregnancies and abortions. These may include limited access to contraceptives or financial or existential insecurity, which may cause parents to decide not to have a child. By identifying these causes, the current government could take steps to eliminate the causes rather than forcibly eliminating the consequences. Qualitative research could be used to examine the level of awareness in different regions about the risks of sex and contraceptive methods, as well as what other options are available besides abortion. In addition, it may be worth assessing why parents do not consider keeping their child to be a good decision: what fears, uncertainties, and variables are behind this choice. By identifying these, the state can provide real help in reducing the number of abortions. What's more the research can serve as a basis for several NGOs and advocacy organizations (e.g., organizations dealing with women, women's rights, and families) to reinforce their efforts with other research and figures.

7. References

[1] Ahmed, S., & Ray, R. (2014). Determinants of pregnancy and induced and spontaneous abortion in a jointly determined framework: evidence from a country-wide, district-level household survey in India. *Journal of biosocial science*, *46*(4), 480-517.

[2] Blank, R. M., George, C. C., & London, R. A. (1996). State abortion rates the impact of policies, providers, politics, demographics, and economic environment. *Journal of health economics*, *15*(5), 513-553.

[3] Caruana-Finkel, L. (2020). Abortion in the time of COVID-19: perspectives from Malta. *Sexual and Reproductive Health Matters*, *28*(1), 1780679.

[4] Čihák, J., Kotrba, V., & Borošová, M. (2025). Determinants of induced abortion rates in Europe. *International Journal of Health Governance*, *30*(3), 320-335.

[5] Csaba, L., & Tünde, F. (2024). Strategic Elements of the Hungarian Family Support System after 2010: Achievements and Challenges. *Európai Tükör*, *27*(1), 195-219.

[6] Dehejia, R., & Lleras-Muney, A. (2004). Booms, busts, and babies' health. *The Quarterly journal of economics*, *119*(3), 1091-1130.

[7] Gemmill, A., Franks, A. M., Anjur-Dietrich, S., Ozinsky, A., Arbour, D., Stuart, E. A., ... & Bell, S. O. (2025). US abortion bans and infant mortality. *JAMA*, *333*(15), 1315-1323.

[8] Gonzalez, F., & Quast, T. (2022). The relationship between abortion rates and economic fluctuations. *Economics & Human Biology*, *46*, 101120.

[9] Guillaume, A., Rossier, C., Translated by Reeve, P. (2018). Abortion Around the World. An Overview of Legislation, Measures, Trends, And Consequences. Population, 73(2), 217-306.

[10] Kulczycki, A. (2023). Between a rock and a hard place: Abortion, catholicism, the populist right and public health threats in Poland. *Religions*, *14*(10), 1271.

[11] Lantz, P. M., Michelmore, K., Moniz, M. H., Mmeje, O., Axinn, W. G., & SPECTOR-BAGDADY, K. A. Y. T. E. (2023). Abortion Policy in the United States: The New Legal Landscape and Its Threats to Health and Socioeconomic Well-Being. *The Milbank Quarterly*, *101*(Suppl 1), 283.

[12] Minkus, Lara and Drobnič, Sonja. (2021) Abortion: Life-Course Stages and Disruptive Life Events. *Zeitschrift für Soziologie*, vol. 50, no. 3-4, 2021, pp. 259-273.

[13] Moreau, C., Shankar, M., Glasier, A., Cameron, S., & Gemzell-Danielsson, K. (2021). Abortion regulation in Europe in the era of COVID-19: a spectrum of policy responses. *BMJ sexual & reproductive health*, *47*(4), e14-e14.

[14] Nagy, B. M. (2023). A terhességmegszakítás jogi szabályozása és az abortuszstigma.
[15] Pető, A., & Svégel, F. (2024). Nationalism, Pronatalism, and the Guild of Gynecology: The Complex Legacy of Abortion Regulation in Hungary. *Central European History*, *57*(4), 499-516.
[16] Simó, F. Z. (2020). Right to Life in Hungary and in the EU: the Ever-Troublesome Issue of Abortion. *Különleges Bánásmód-Interdiszciplináris folyóirat*, *6*(4), 83-90.
[17] Srinivasulu, S., & Heiland, F. W. (2025). How will abortion bans affect maternal health? Forecasting the maternal mortality and morbidity consequences of banning abortion in 14 US States. *Journal of Women's Health*, *34*(6), 843-854.
[18] SZABÓ-MORVAI, A. W. Á. (2023). Precautionary Fertility: Conceptions, Births, and Abortions around Employment Shocks.
[19] Szalma, I., & Sipos, A. (2024). A comparative analysis across reproduction policy fields in Hungary. In *Reproduction Policy in the Twenty-First Century* (pp. 119-135). Edward Elgar Publishing.
[20] Wesselink, A. K., Wise, L. A., Hatch, E. E., Mikkelsen, E. M., Savitz, D. A., Kirwa, K., & Rothman, K. J. (2022). A prospective cohort study of seasonal variation in spontaneous abortion. *Epidemiology*, *33*(3), 441-448.